Improve Generalization

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Improve Generalization

> Prepare Improve Generalization: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Compare **training** and validation **evidence** for model **capacity**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/16, ai-4/27, ai-4/30. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

Look for: Create features available at **prediction** time

Reference example, not the only acceptable wording: Teacher explanation: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

Overfitting

Without opening the old lesson, explain one key idea from **Overfitting**. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model **capacity**

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Look for: Select a model using a declared validation rule

Reference example, not the only acceptable wording: Teacher explanation: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite **training** MAE 0.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

Concept to explain

Different failure causes require different repairs: simplicity, representative data, regularization, or better features.

Term	Meaning
generalization	performance beyond training examples
controlled change	one planned modification
ablation	remove one element to measure its effect
regression test	protect previously passing behavior

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **overfitting** and model **capacity**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

No; it may fit peculiarities of the **training** sample.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Apply the connected idea: Distribution shift

Teach two 45-minute sessions. This week focuses on **distribution shift**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Performance may differ when **input**s, labels or conditions change.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain improve **generalization** using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Improve Generalization using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Choose one controlled repair and rerun the same evaluation.

MODEL IDEA Different failure causes require different repairs: simplicity, representative data, regularization, or better features.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD **Evidence**: high train score, low validation score, largest gap in low light. Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 **Evidence**: high train score, low validation score, largest gap in low light.

Step 2

STEP 2 Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.

Step 3

STEP 3 Add representative low-light **training** examples while freezing model settings.

Step 4

STEP 4 Rerun all validation cases; compare overall, error slice, subgroup gap, and regressions.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

A classifier relies on tray colour; new trays reduce performance.

The teaching case

Propose controlled improvement.

Exact explanation to model

Why the result follows

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

Explain the added worked example

Explain the added worked example · 1

A more flexible model can represent complicated relationships and accidental details. **Capacity** can help when there is enough relevant **evidence**, but perfect **training** performance is not the goal.

Compare **training** and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, **reference labels** and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **overfitting** and model **capacity**.

Explain the added worked example · 2

On the same regression split, a constant has **training** MAE 5 and validation 6, a line has **training** 2 and validation 3, and a highly flexible curve has **training** 0 and validation 8. Under smallest validation MAE, choose the line.

The curve has the lowest **training** loss but the worst validation result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Capacity	Flexibility of a model to represent patterns
Overfitting	Learning details that fail to generalize well
Underfitting	Failing to capture useful structure
Generalization	Performance on relevant unseen examples

Apply the connected idea: Distribution shift

Apply the connected idea: Distribution shift · 3

A model trained on one set of conditions may encounter another. Lighting, measuring tools, class frequency and the relationship between **input**s and labels can change.

A shift does not automatically imply failure, but it challenges the assumption that old **evidence** represents current use.

Use fictional classroom data throughout this investigation. Keep the task, units, **reference labels** and data split visible as you reason. Earlier weeks supplied the model-building **process**; this week examines **distribution shift**.

Apply the connected idea: Distribution shift · 4

On development-like bright images, a frozen classroom classifier makes 2 errors in 20 cases:10 percent. On a separate dim set it makes 6 errors in 20:30 percent. The pooled result is 8/40=20 percent error.

The pooled score hides a 20-percentage-point difference between conditions.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Guided response: It combines conditions and hides the higher observed dim-condition error.

Exit response: Flag unsupported **input**s, pause the affected model decision, use the declared human or non-model **fallback**, and gather appropriate **evidence** before revision.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Overfitting and model capacity at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Improve Generalization

Test whether a plant model follows the leaf or a convenient background shortcut.

The facts we will use

Supplied facts

Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards. Intended use allows both backgrounds for either class.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/30

Improve Generalization

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Notice correlation

Background colour perfectly tracks the labels in this training sample.

2. Design crossed cases

Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.

3. Inspect fixed model

Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

CHANGE ONE THING

The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Notice correlation	Background colour perfectly tracks the labels in this training sample.
2. Design crossed cases	Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.
3. Inspect fixed model	Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

Change one thing in the pictured case

Change one thing

Change: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

Predicted result

Prediction: It fails on spotted/grey and healthy/red in the crossed audit.

Why the result follows

Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Improve Generalization, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: A classifier relies on tray colour; new trays reduce performance.

2. State the decision to explain

Write the question to explain: Propose controlled improvement.

3. Reveal the worked reasoning

Reveal after **prediction**s: Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

Hold back the independent task

Keep for an independent attempt: A team proposes more representative **training** examples and a simpler model. How should it compare changes while protecting the final test?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: No. It is consistent with **overfitting** but may also reflect shift, data quality or split problems.

Exit response: Check data provenance, split grouping, label quality and repeated development results; inspect representative residuals without using final test feedback to tune.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Can a zero training error guarantee good validation performance? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify capacity in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: select from evidence; interpret a gap. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between capacity and overfitting. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Describe one fair next development comparison.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Compare training and validation evidence for model capacity; Explain overfitting without assuming every gap proves it. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Teach the connected idea

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Why does a test score need a population description? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify distribution shift in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: compare conditions; read pooled evidence. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between distribution shift and monitoring. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Can changed predictions alone prove an accuracy drop?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Recognize changes between development and later data; Compare errors across conditions with denominators. Record any support used.

Stage: 35 to 45 minutes

> **Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion practice

Completion task: Does a **training**-validation gap alone prove **overfitting**?

Reasoning to listen for

Teacher check: Teacher explanation: No. It is consistent with **overfitting** but may also reflect shift, data quality or split problems.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Distribution shift

Target: Recognize changes between development and later data

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

> Completion practice: Why is 8/40 incomplete by itself?

> **Reasoning to listen for**
>
> Teacher check: Teacher explanation: It combines conditions and hides the higher observed dim-condition error.

Individual check: For **Distribution shift**, use its worked case to evaluate this proposed explanation: “A final answer is enough”. Explain the deciding step.

> **Expected repair**
>
> Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Bright 2/20=10 percent; dim 6/20=30 percent; dim is 20 percentage points higher.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Freeze the revised model and score new grouped crossed-background examples; reused audit cases remain development **evidence**. Changed-case check: It fails on spotted/grey and healthy/red in the crossed audit.

Those supplied **prediction**s follow background rather than the intended leaf target.

Changed-case reasoning: It fails on spotted/grey and healthy/red in the crossed audit. Those supplied **prediction**s follow background rather than the intended leaf target.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Improve Generalization investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Repair Planner: match four **evidence** patterns to targeted changes, run one change, and explain what the result does not prove.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Changing data, features, model, and threshold together is the fastest proof of improvement.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

A stronger claim makes a stronger project: Check data provenance, split grouping, label quality and repeated development results; inspect representative residuals without using final test feedback to tune.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: B wins the validation tie by the declared simplicity rule.

C is not selected merely for its zero **training** error.

Check these misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Bright 2/20=10 percent; dim 6/20=30 percent; dim is 20 percentage points higher.

A stronger claim makes a stronger project: Flag unsupported **input**s, pause the affected model decision, use the declared human or non-model **fallback**, and gather appropriate **evidence** before revision.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Rates 10 percent and 50 percent; pooled 8/40=20 percent.

Report both denominators and the 40-point condition gap.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Improve Generalization questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

Independent question

Independent question · withhold the key

A team proposes more representative **training** examples and a simpler model. How should it

compare changes while protecting the final test?

Expected independent answer

> **Teacher key · Expected independent answer**
>
> Use the same validation protocol and change one factor at a time where possible. Record results and retain the final test for the fixed final choice.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Repeatedly checking final test scores is the best way to choose every improvement.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Collect both classes on multiple backgrounds with reliable labels. Compare a planned **training**-data change on fixed validation slices, including changed backgrounds, and inspect errors.

Keep leaf identity partitions and final test sealed until selection.

Explanations for the additional workbook examples

W1 Select from evidence

Workbook question · W1

Choose among the three worked candidates and explain the **training** trap.

Teacher reasoning · W1

Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

W2 Interpret a gap

Workbook question · W2

Does a **training**-validation gap alone prove **overfitting**?

Teacher reasoning · W2

Teacher explanation: No. It is consistent with **overfitting** but may also reflect shift, data quality or split problems.

W3 Propose a controlled change

Workbook question · W3

Describe one fair next development comparison.

Teacher reasoning · W3

Teacher explanation: Compare a simpler curve against the current curve on the same validation rows, retaining the same metric and **training**-only **fitting**; log the change.

W4 Apply independently

Workbook question · W4

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

Teacher reasoning · W4

Teacher explanation: B wins the validation tie by the declared simplicity rule. C is not selected merely for its zero **training** error.

W5 Limit a diagnosis

Workbook question · W5

What extra **evidence** would you inspect before concluding a model memorized noise?

Teacher reasoning · W5

Teacher explanation: Check data provenance, split grouping, label quality and repeated development results; inspect representative residuals without using final test feedback to tune.

Distribution shift — W1 Compare conditions

Calculate each worked error rate and the difference.

Teacher reasoning

Teacher explanation: Bright 2/20=10 percent; dim 6/20=30 percent; dim is 20 percentage points higher.

Distribution shift — W2 Read pooled evidence

Why is 8/40 incomplete by itself?

Teacher reasoning

Teacher explanation: It combines conditions and hides the higher observed dim-condition error.

Distribution shift — W3 Wait for labels

Can changed **prediction**s alone prove an accuracy drop?

Teacher reasoning

Teacher explanation: No. **Prediction** patterns can change without known correctness; valid **reference labels** are needed to measure error.

Distribution shift — W4 Apply independently

Old-like data has 3 errors/30 and new-condition data 5/10. Give both rates and the pooled rate.

Teacher reasoning

Teacher explanation: Rates 10 percent and 50 percent; pooled 8/40=20 percent. Report both denominators and the 40-point condition gap.

Distribution shift — W5 Plan a response

State a safe classroom response to **input**s outside the supported range.

Teacher reasoning

Teacher explanation: Flag unsupported **input**s, pause the affected model decision, use the declared human or non-model **fallback**, and gather appropriate **evidence** before revision.

Assess independent transfer

Independent question · withhold the key

Selected case: Spotted leaves have blue backgrounds and healthy leaves white backgrounds in **training**. Design a generalisation investigation with fresh evaluation.

Selected case answer

> **Teacher key · Selected case answer**
>
> Cross background and class conditions; compare the same leaf on changed backgrounds as a diagnostic and group views by plant in splits.
>
> Use development data for decisions, then compare frozen versions on the same untouched final records. Record actual outcomes; the correlated backgrounds alone do not prove a shortcut.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Improve Generalization to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Improve Plant Classifier with one approved change and a full before/after **evidence** table.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 31: Subgroup Performance. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **capacity** and **overfitting** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: B wins the validation tie by the declared simplicity rule. C is not selected merely for its zero **training** error.

Fresh retry **training**/validation D 3/4 E 0/7 F 2/3: choose F under lowest validation loss. Explain why E is not automatically best. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Learning curves and variability. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Readiness and reassessment for Distribution shift

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **distribution shift** and **monitoring** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Rates 10 percent and 50 percent; pooled 8/40=20 percent. Report both denominators and the 40-point condition gap.

Fresh retry condition A 1/10 errors and B 3/10:10 and 30 percent, pooled 4/20=20 percent. Require separate counts and no causal claim from the table alone. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Plan the delayed return

Suggested return points in this level: ai-4/31, ai-4/33, ai-4/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	

Record	Teacher evidence
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Improve Generalization lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Compare **training** and validation **evidence** for model **capacity**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Does a **training**-validation gap alone prove **overfitting**?

Supported practice answer

Reasoned practice answer: Teacher explanation: No. It is consistent with **overfitting** but may also reflect shift, data quality or split problems.

Check every learner before discussion: Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Require a **prediction** and a reason before comparing.

Expected result: It fails on spotted/grey and healthy/red in the crossed audit. Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

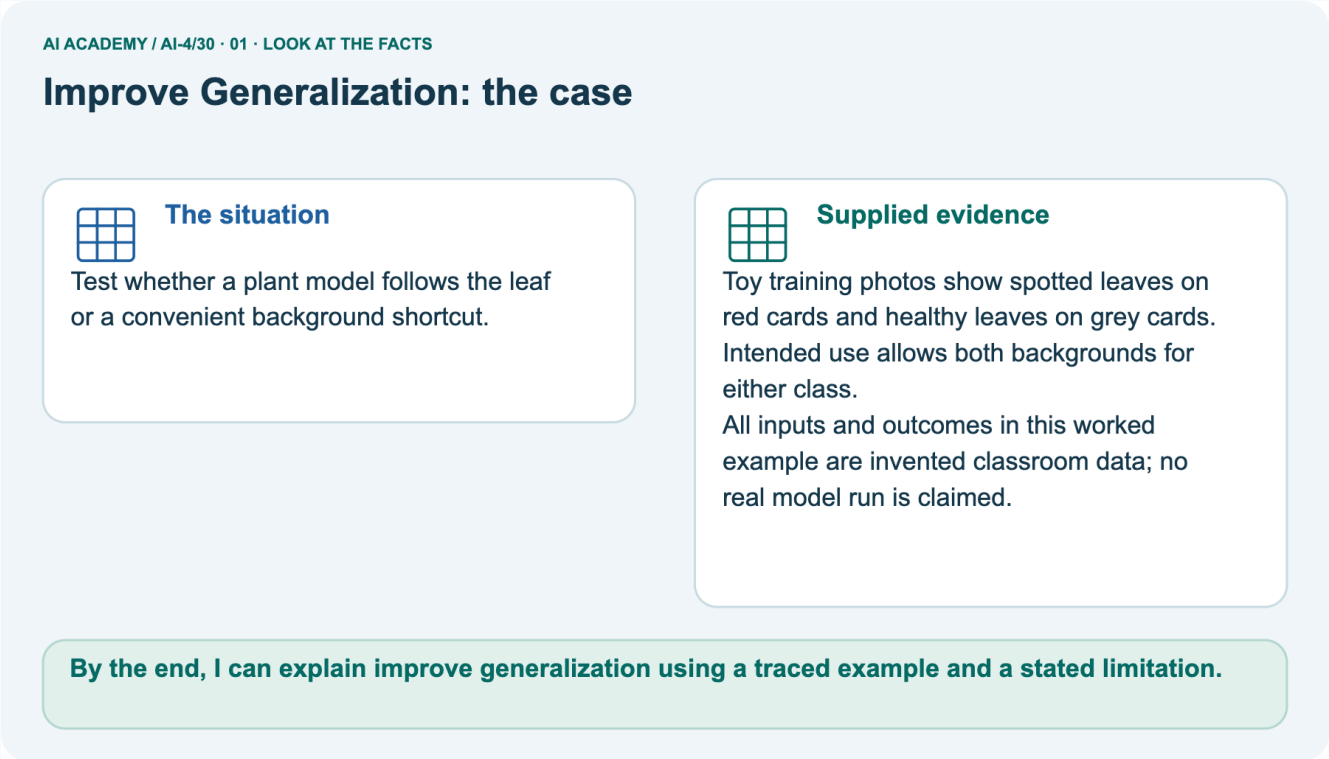
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Improve Generalization graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Improve Generalization: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Test whether a plant model follows the leaf or a convenient background shortcut. Supplied facts: Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards.

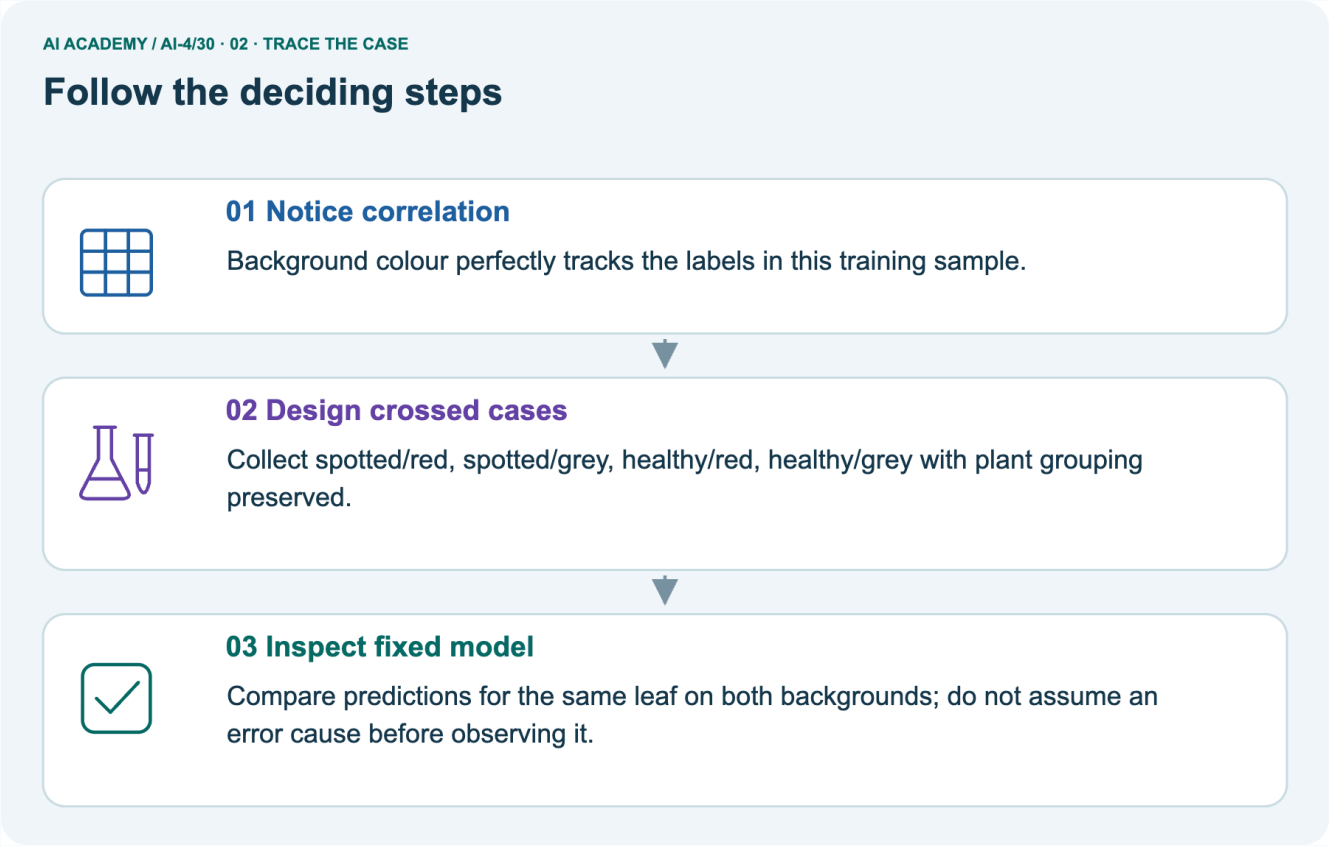
Intended use allows both backgrounds for either class. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Notice correlation: Background colour perfectly tracks the labels in this **training** sample. → Design crossed cases: Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved. → Inspect fixed model: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

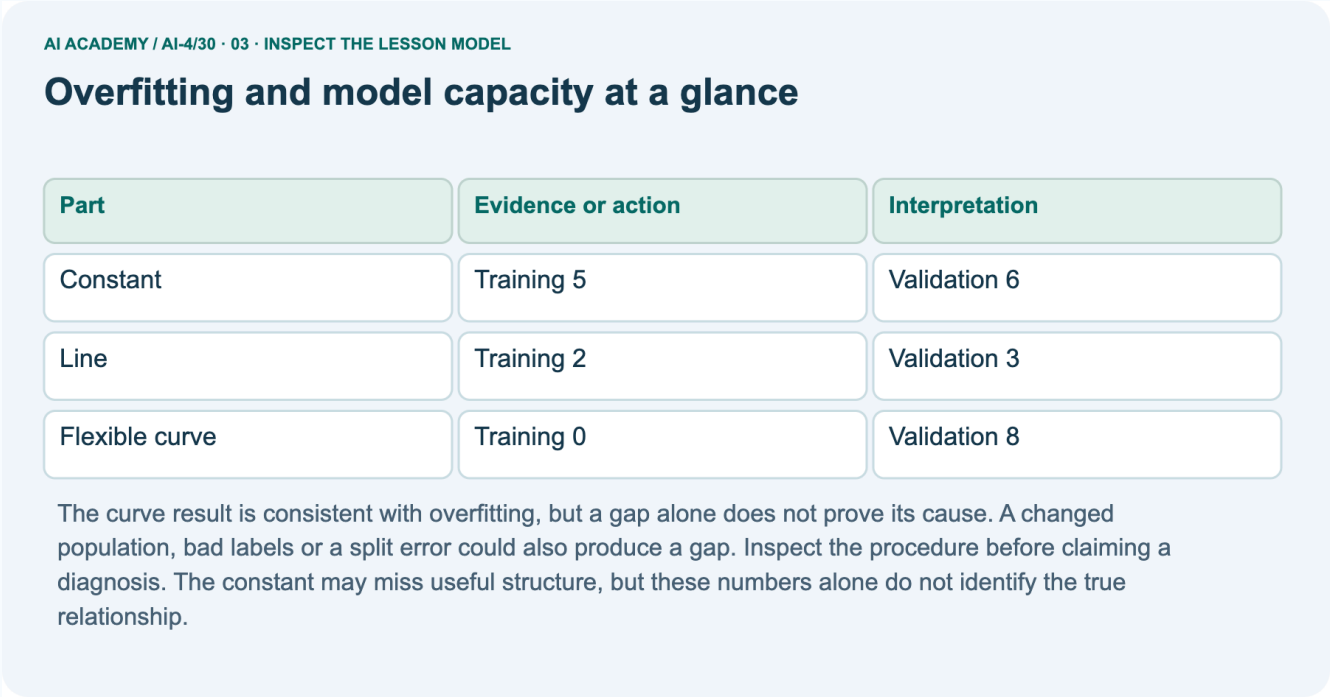
> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if

the image is difficult to see.

03. Overfitting and model capacity at a glance



Follow the labelled picture with the matching explanation. Open the vector diagram

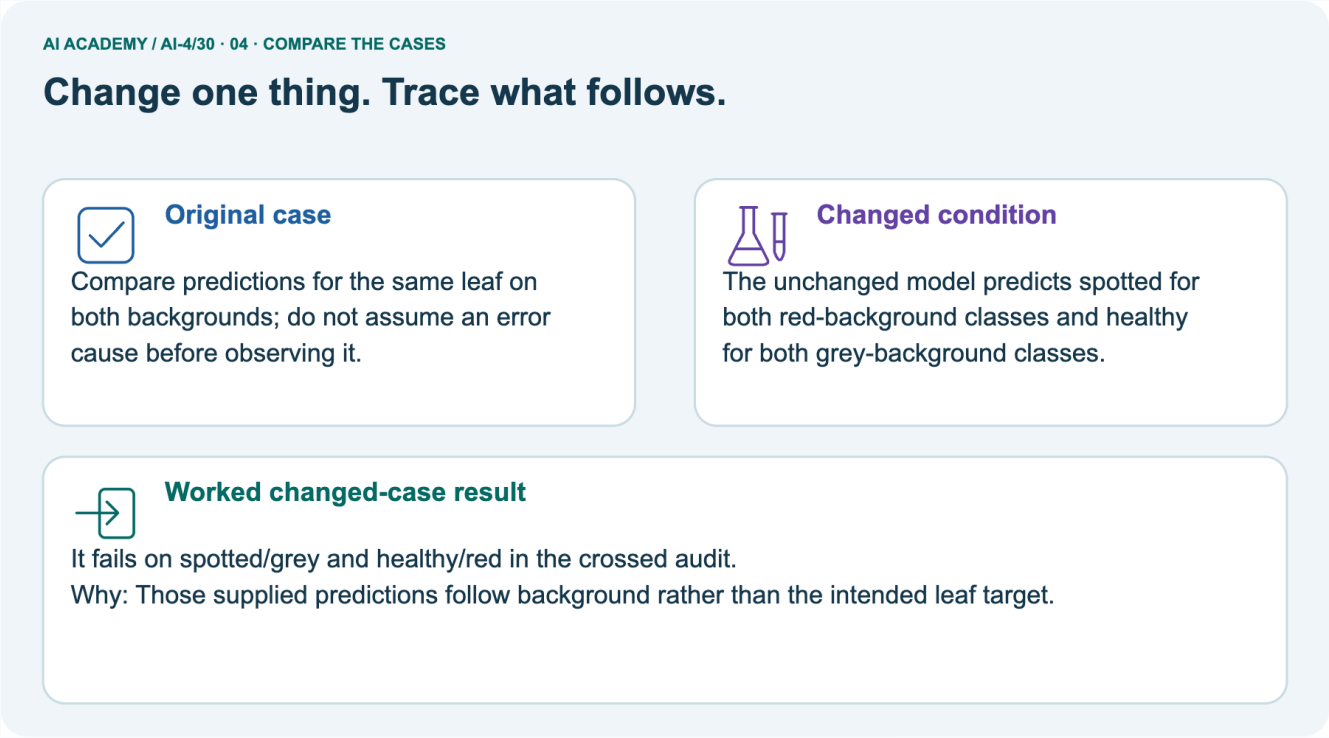
Overfitting and model **capacity** at a glance. Constant | **Training** 5 | Validation 6. Line | **Training** 2 | Validation 3. Flexible curve | **Training** 0 | Validation 8

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

Changed condition: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Expected result: It fails on spotted/grey and healthy/red in the crossed audit.

Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

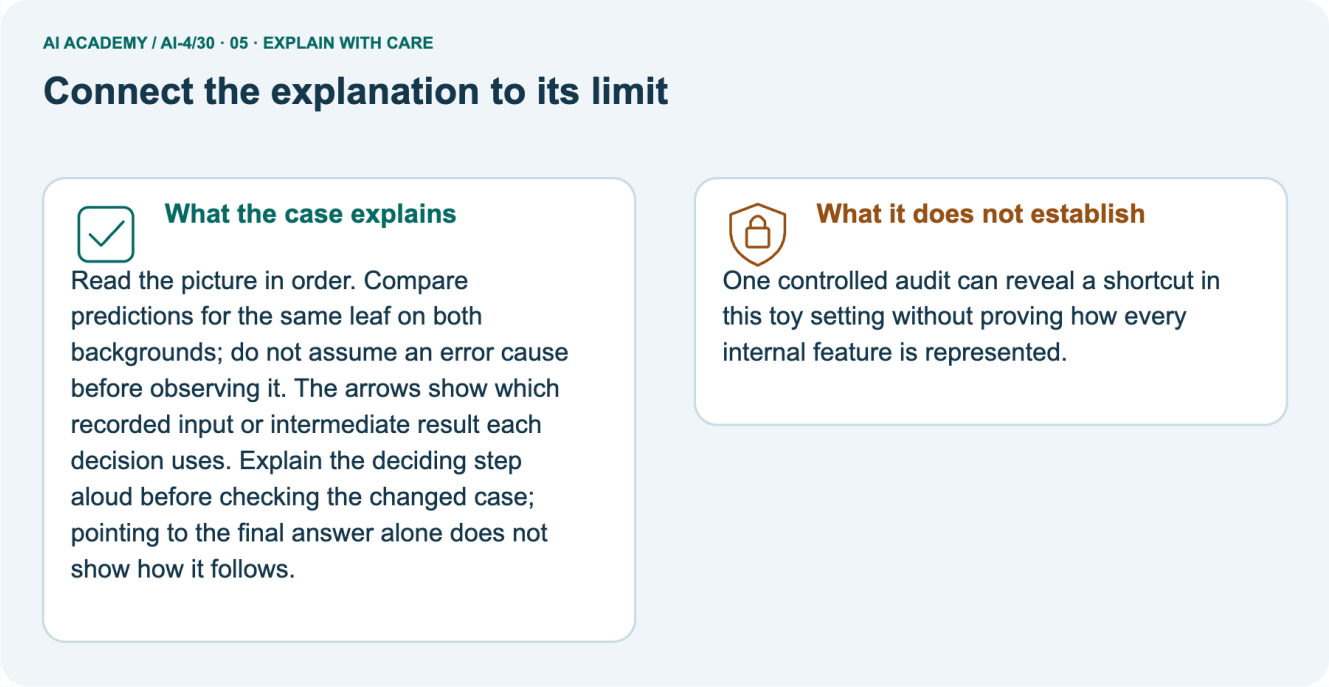
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Those supplied **prediction**s follow background rather than the intended leaf target.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

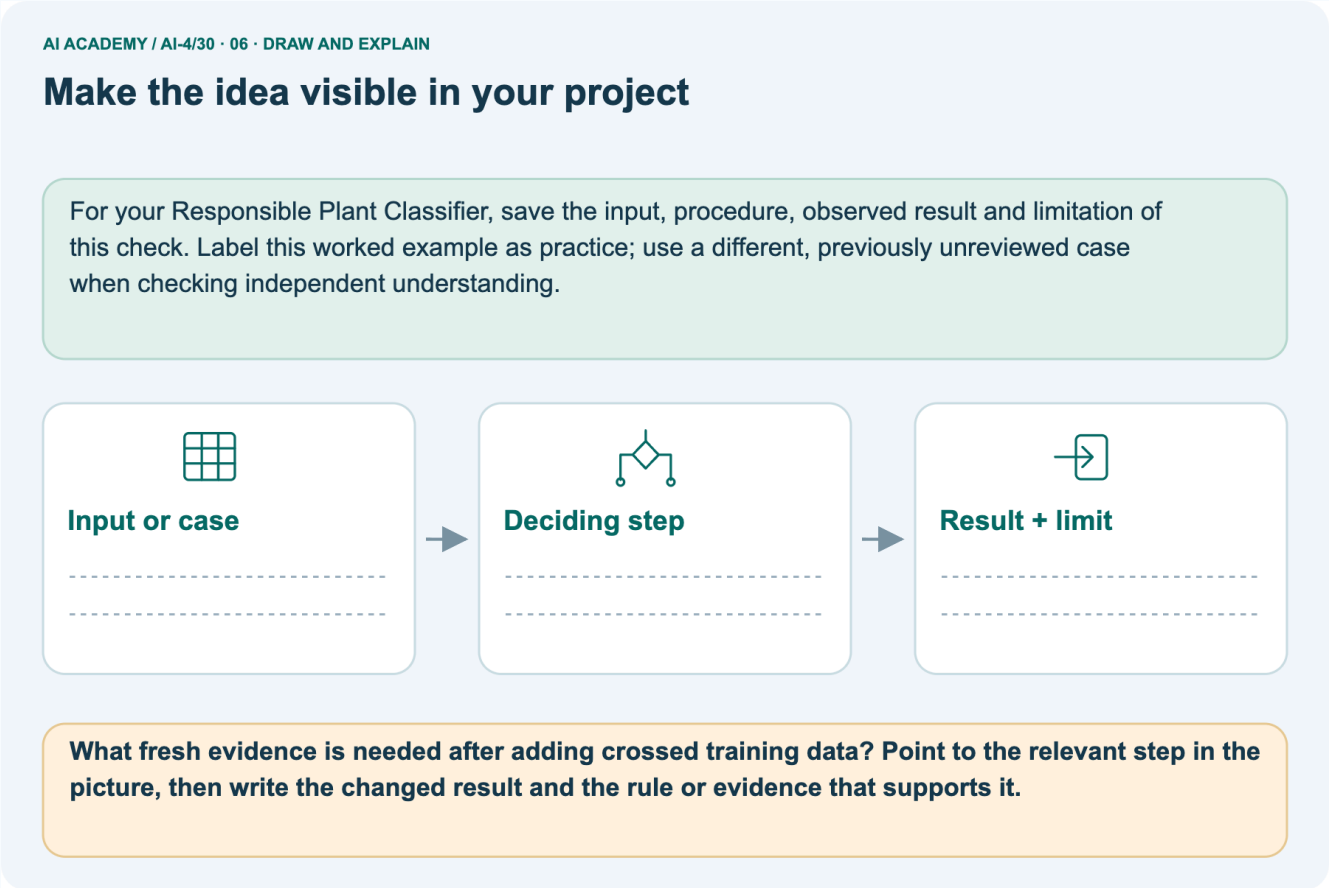
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Freeze the revised model and score new grouped crossed-background examples; reused audit cases remain development **evidence**.

Changed-case check: It fails on spotted/grey and healthy/red in the crossed audit. Those supplied **prediction**s follow background rather than the intended leaf target.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.